

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Algebraic Roots & Indices

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ALGEBRAIC ROOTS & INDICES · CH2 · L2



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The Map

This lesson collapses Algebraic Roots & Indices into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use index laws with letters

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use negative powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use fractional powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Simplify before substituting

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Simplify surd

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 49 website questions, 49 checked solutions, 3 local PDFs read.

01 Use index laws with letters

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt{12} = 2\sqrt{3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q17

Simplify surd

$$\sqrt{12} = 2\sqrt{3}$$

Simplify surd

$$6 + 2\sqrt{12} = 6 + 4\sqrt{3}$$

Finish: (a) shown, (b) $\frac{t^{10}}{9a^8}$

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

17 (a) Show that $6 \frac{2}{3} \times 12 \frac{1}{2} \times 7 \frac{4}{5} \times 3 \frac{2}{3} + () = + ()$ Show each stage of your working. (3) (b) Simplify fully $27 \frac{1}{2} \times 15 \frac{2}{3} \times 3 \frac{1}{4} \times 2 \frac{3}{4}$ (3)

YOUR SOLUTION

02 Use negative powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{x^9}{x^2} = x^{9-2} = x^7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q1

(a)

$$\frac{x^9}{x^2} = x^{9-2} = x^7$$

(b)

$$\frac{7^8 \times 7^4}{7^3} = 7^{8+4-3} = 7^9$$

Finish: (a) x^7 , (b) 7^9 .

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use fractional powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q21

Factorise expression

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

Use index laws

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)}$$

Finish: (a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$.

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

21 (a) Simplify fully $10^3 23 12 4 9 2 2 \times x \times x + + - (3) 2 2 8 4 2 3 2 5 y y y n$ times = + (b)
Find an expression for n in terms of y. Show clear algebraic working and simplify your expression. (4)

YOUR SOLUTION

04 Simplify before substituting

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt{12} = 2\sqrt{3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q17

Simplify surd

$$\sqrt{12} = 2\sqrt{3}$$

Simplify surd

$$6 + 2\sqrt{12} = 6 + 4\sqrt{3}$$

Finish: (a) shown, (b) $\frac{t^{10}}{9a^8}$

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

17 (a) Show that $6 \frac{2}{3} \times 12 \frac{1}{2} \times 7 \frac{4}{5} \times 3 \frac{2}{3} + () = + ()$ Show each stage of your working. (3) (b) Simplify fully $27 \frac{1}{2} \times 15 \frac{2}{3} \times 3 \frac{1}{4} \times 2 \frac{3}{4}$ a t - (3)

YOUR SOLUTION

05 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q21

Factorise expression

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

Use index laws

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)}$$

Finish: (a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$.

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Simplify surd

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt{12} = 2\sqrt{3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q17

Simplify surd

$$\sqrt{12} = 2\sqrt{3}$$

Simplify surd

$$6 + 2\sqrt{12} = 6 + 4\sqrt{3}$$

Finish: (a) shown, (b) $\frac{t^{10}}{9a^8}$

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

17 (a) Show that $6 \frac{2}{3} \times 12 \frac{12}{13} \times 4 \frac{3}{4} \times 2 + () = + ()$ Show each stage of your working. (3) (b) Simplify fully $27 \frac{12}{15} \times 2 \frac{3}{4} \times t - (3)$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use index laws with letters
"work out", "show", "hence", a familiar exam phrase	Use negative powers
"work out", "show", "hence", a familiar exam phrase	Use fractional powers
"work out", "show", "hence", a familiar exam phrase	Simplify before substituting
"work out", "show", "hence", a familiar exam phrase	Use index laws
"work out", "show", "hence", a familiar exam phrase	Simplify surd

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.